Guide to the Project manaGement toolkit

Tools and techniques for managing Local Government projects

NSW & ACT
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INSTITUTE OF PUBLIC WORKS
ENGINEERING AUSTRALASIA

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Contents

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Important Note to Readers

Any legislation cited in this Guide is a reference to NSW legislation in force at the time of this publication unless otherwise noted.

For further information contact:

Institute of Public Works Engineering Australasia (NSW Division) Ltd
447 Kent Street SYDNEY 2000
Phone: 02 8267 3001
Email: admin@ipweansw.org

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Introduction

Who Should Use this Guide?

This Guide has been developed to be used by non-professional project managers but it does not preclude its use by skilled and highly experienced project managers. The resource has been developed for councils where project management skills and knowledge are lacking or resources are under pressure. The Guide and the templates can be used to complement existing project management systems or provide consistent, quality assured approaches where systems do not currently exist.

How to Use the Guide

This Guide will help you to use the tools and techniques to manage a small to medium project. It is divided into four phases based around four stages in the project life cycle. Each phase has a brief explanation of why this stage is important and at the end of the section is an Action icon outlining what you should do at this stage of your project and the tools that will assist you. There are five folders of templates, one for each phase of the project and one of checklists. There is also a checklist for each project phase and an A3 wall poster summary.

Follow the Action Icon

Where you see this icon, you will find a list of project management tools and templates that will assist you with your project. You will find blank templates as well as some completed samples. You do not have to use all the tools and there are various options to choose from. Just select the ones you are most comfortable to use and meet your requirements. You can also add to the templates if you need for fields to meet council requirements. You can add your logo, council name, page numbers and document numbering as well.

Some of the tools contain data. This is easily replaced but will give you some idea of how the tool works. Excel spreadsheets generally self-populate when you change data in the cells.

What is a Project?

Projects can be of any size and duration. They can be as simple as planning a team building day or as complex as redeveloping a town centre or organising a major event. A local government project can be funded from a number of sources including capital, maintenance, or operational funds from Council or external sources such as Government Grants or a combination of these. Regardless of the funding source all need the same amount of planning to be successful.

Projects:

- are a temporary endeavour undertaken to achieve a particular outcome
- have a definite beginning and end
- consist of managed and controlled actions
- are required to meet specific objectives within time, cost and quality constraints

What is Project Management?

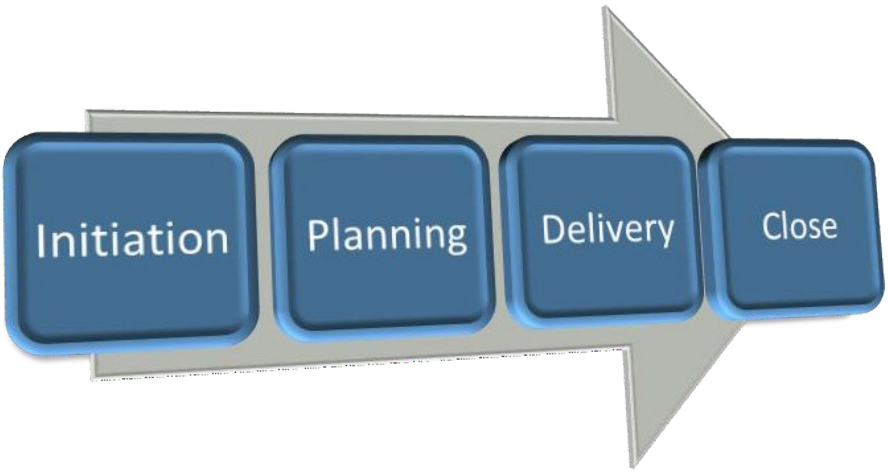
Project management is a critical practice that applies knowledge, skills, tools and techniques to a broad range of activities in order to meet the requirements of the particular project.

Project management involves planning, control and co-ordination of a project from inception to completion, ensuring timeframes, budget, administrative obligations and quality are met.

Project management brings together skills that relate to the management of time, cost, quality, scope, risk and human resources among others.

What is the Project Lifecycle?

The project life cycle is made up of five stages: initiation, planning, execution, monitoring and control and closure. In this kit the execution, monitoring and controlling phases have been combined under the heading delivery.



The diagram below provides an indicative summary of the phases in the project life cycle and the key elements of project management.

Key Elements	Phase 1 - Initiation	Phase 2 - Planning	Phase 3 - Delivery	Phase 4 - Close
1. Justification, Objectives and Drivers	X	X	X	X
2. Scoping	X	X	X	
3. Governance	X	X	X	X
4. Stakeholder Engagement	X	X	X	X
5. Risk Management	X	X	X	X
6. Issues Management			X	X
7. Cost Estimation, resourcing and Procurement	X	X	X	X
8. Scheduling	X	X	X	X
9. Quality Management			X	X
10. Reporting			X	X
11. Implementation Review and Evaluation			X	X

Must Have Project Management Skills

To succeed as a project manager, you will need a combination of technical and behavioural skills.

Planning: As the project manager you're in charge of creating the roadmap that will see the project delivered successfully and to a high quality. As part of your project plan you will set the project goals and the scope of the project. An essential part of project planning is the creation of a good business plan that outlines the benefits of the project, which can then be used to assess its efficiency and effectiveness. The business case will even go as far as identifying whether the project is worth pursuing in the first place.

Scheduling: Another technical skill in the initial stages of the project is scheduling, which outlines the tasks to be completed and designates responsibility for the varying deliverables to the project. A Gantt Chart is a popular tool for outlining the project schedule, however there are also plenty of project management software applications available, which helps streamline the process of project scheduling and will keep you on track with your project deliverables.

Budgeting: When you're creating a budget for your project, be realistic about the timeline that the project can be completed and about how much each deliverable will cost. Overpromising or an incorrect project budget will result in greater stress as you near the finish line of the project and will almost always end up seeing you run over the initial budget.

Risk management: Another key project management skill is being able to identify risk before a project commences and mitigate those risks to the best of your ability. Conducting a risk assessment will highlight any risk events that could potentially occur, allow you to calculate the risk of that event taking place and its impact and provide you with the opportunity to plan out how you can treat it if it does occur.

Contract management: Depending on the organisation you are working for, contract management may fall into your remit as a project manager or it may be the task of someone reporting to you. Even if you're not the individual drawing up the contract in your organisation, ensure you're aware of what the contract covers. As a project manager you should have a thorough understanding of the terminology specific to the industry you're working in and have an understanding of legislation and common law.

Communication: You've probably heard this before – honing your behavioural skills is just as important as having great technical skills. While there have been discussions of the digitalisation of the project management process due to artificial intelligence, the behavioural skillset is impossible to program or replicate by machines and is continuously growing in demand in the marketplace. As the project manager, good communication skills will see you keep all parties to a project informed and up to date with the progress of the job. It is vital that the people/organisations/community groups that need to be informed during the course of the project are identified early in the process. This includes other divisions or work teams from within your Council.

Leadership: What happens when something goes wrong on a project? Whether it's the client, contractors or senior members of the organisation, the project manager is the key person that people turned to for a solution. This is where having effective leadership skills comes into play. As a leader being consistent and reliable will win the trust of your team. If leadership isn't your strong suit, take the time to develop your emotional intelligence by undertaking a leadership course. "Leadership is about setting the benchmark high, holding yourself accountable to that, and giving people every support so you can help people meet that benchmark," Lisa Harrison, Capability Consultant

Motivation: Another behavioural skill that should be part of a project manager's skillset is the ability to motivate your team. Staff retention is a key factor for managers, so having a project manager who is able to establish a climate and culture of trust and collaboration is very important to decision-makers. The project manager with a heightened sense of awareness to what motivates their team, how best to utilise strengths and weaknesses, as well as the best ways to reward teams are invaluable assets to any organisation.

Networking: You've probably heard the saying "it's not what you know but who you know". A highly sought after project manager will be engaged with both their industry and their profession through project management networking. By attending project management events and connecting with other project managers it could result in your next client or contract and is the perfect way to keep you up to date on what is happening in your industry and profession.

Source: https://www.aipm.com.au/project-management-career-path/skills (March 2021)

Phase 1 Initiation

Project Justification : Projects are usually justified in terms of meeting council or funding agreement objectives and should be closely aligned to them. This alignment is explored through initial scoping which culminates in the preparation of a Business Case, which should:

- explore the underlying business drivers
- describe the relationship of the proposed project to Council's strategic priorities
- define the relative priority assigned to the project
- analyse the capability and capacity of the organisation to undertake the project and absorb the change the project bring about
- identify 'critical success factors' including, but not limited to time, budget and/or quality criteria.

Should we do this project: Ultimately the business case should answer the question: "Should council proceed with this project?" and should explain why. Before proceeding, the appropriate decision-making body or person needs to endorse the project should proceed.

Once the project has been endorsed, the Business Case may be used as the basis for an application for funding from the NSW Government, community consultation, additions to council plans, the development of specifications for procurement of contracted services and the allocation of finances to the project.

The Project Manager should be involved in the initiation stage of a project but sometimes the project may have been initiated by others and some of the scoping and business planning already commenced.

Applying for funding can be a project in itself and the initiation tools can be used to develop the case and information required for a funding application. It is essential to cost the project as accurately as possible at this stage to ensure sufficient funds are applied for.

Project Business Case

All projects require a document that demonstrates due diligence by council in project scoping, planning, resourcing and governance: The business case template provided:

- summarises the project and background and provides the key project milestones
- recognises the key milestones of Council or other agency approvals needed under the Local Government Act 1993 or other legislation
- demonstrates adequate stakeholder engagement and support
- demonstrates capacity and governance systems are in place to deliver the project within scope, specification and budget
- forecasts likely expenditure in a way that is suitable for incorporation in the funding deeds and/or Council's financial accounting and reporting system.
- confirms council's commitment to the project and acceptance of ongoing ownership obligations of any new assets arising.

Council also needs, as a minimum, a simplified version of the business case for its own reasons. Infrastructure projects, especially those on a large scale undertaken by councils can pose a significant financial and management risk to councils. Preparing a business case will provide the following benefits:

- The plan documents the reasons for needing to expend council funds on a solution (environmental, public health, economic development) for endorsement by the elected council.
- The plan demonstrates that the proposal is consistent with strategic planning processes, in particular the Integrated Planning and Reporting framework mandated for NSW councils. (ss. 402 to 406 of the Local Government Act).
- Councils can ensure elements of the Integrated Planning and Reporting framework are specifically addressed. This includes: alignment with the aspirations of the community in the community strategic plan; identification of need in the asset management plan; demonstrating that council can afford to undertake the works through scenario modelling in the long-term financial plan (which should also include impact on typical residential bills).
- Councils can demonstrate (to itself and the community) that the project or solution is consistent with its four-year delivery program (or identify that the delivery program will be amended).
- A version of the business case/project plan can help satisfy audit and risk management requirements under s. 428 of the Local Government Act and for the NSW Audit Office.

However, for projects with a total capital investment above $10 million (excluding GST), it is recommended councils use the NSW Treasury 'Detailed business case template' available for download from www.treasury.nsw.gov.au/index.php/information-public-entities/business-cases, or an equivalent document.

A sample business case template is included in the Initiation templates.

Initiating the project means setting up the project from idea (thought bubble) or inception and making sure it is:

- the right project
- in the right place
- at the right time
- for the right purpose.

The initiation phase is essential to capturing:

- an early understanding of the project rationale/the business drivers (including how it links to and contracts and council plans)
- an initial statement of the project objectives
- the high-level project outputs
- the scope of the project
- the potential sponsor of the project
- stakeholders within the council who will be essential to the project's success
- Cost estimates
- Risks and assumptions

When initially planning a project, it is imperative to define the project in terms of the desired benefits (project outcomes) and the products or services that are required to achieve them (project outputs).

In the initiation phase it is important to:

a) Scope the project

b) Identify project approval processes

c) Develop or source a project brief

d) Clarify the roles and responsibilities of the project team and who makes decisions (governance)

e) Identify stakeholders essential to the project

f) Clearly define the deliverables

g) Develop cost estimates

h) Develop a work breakdown structure

For guidance regarding stakeholder analysis and engagement refer to council's community engagement policies. This details how to prepare a project-specific engagement strategy which is required if the project is to involve stakeholder consultation. This is also the time to allocate tasks to other team members or divisions within the council.

Actions – What to Do

Use the Project Initiation Checklist (Checklist 1) and the initiation tools to ensure you have a clear picture of the project and what it aims to achieve before moving onto Phase 2 Planning. Much of what you complete in this stage can be included in the final project plan.

You should complete the following:

- Business Case
- Access the contract or tender documentation to ensure you know what you are required to deliver
- Project Scope
- Work Breakdown Structure
- Initial Cost Estimates

Research indicates that most failed projects have either skipped the project initiation phase altogether or have been through inadequate initiation process. In many cases the more obvious objectives such as "on time" and "on budget" were emphasised at the expense of other more important and often not immediately present objectives. A useful initiation exercise to assist in scoping a project and defining its broad deliverables is to determine what is IN and what is NOT to be included within the bounds of the project deliverables.

Key Elements in the Initiation Phase

1. Should Council be doing this Project?

A business case should demonstrate the case for change; including whether Council has the resources and funds available to operate, maintain and replace the project outputs. The Project Manager may need assistance from others in the organisation (or community) to answer the questions whether it is the right project, in the right place, at the right time and for the right purpose.

2. Funding Agreement and Responsibilities

Projects within council will have a business case that demonstrates the case for change. These projects may be funded by council or through a funding deed with an external funding agency such as Public Works Advisory and Regional NSW. In some cases, the project funds come from the state government but the responsibility for executing the project lies with council. This may be more complicated with an external project manager and assurance manager. These roles need to be incorporated into the governance structure (point 3) and reporting frameworks.

When managing a project, it is essential to obtain a copy of these documents so you can be clear on the project objectives, time frames and funding. If you are not clear on these aspects of the project speak to others who developed the funding submission or business case. This knowledge will help you scope the project.

The initiation tools are useful in collecting the information needed for making funding applications.

3. Scoping

The scope of a project will specify what can be delivered within the timeframe and the resource constraints imposed by the project. Scoping establishes the boundaries of a project and should occur regardless of the size of the project.

Scoping is not a static, one-off process. While initial scoping occurs in Phase 1 - Initiation, the scope of the project will be re-examined many times over in the project's life. In theory, the more complex a project, the more time should be spent undertaking initial scoping activities e.g. detailed feasibility study or cost benefit analysis. However, many projects are initiated based on a brief proposal, a Council resolution or a grant from Federal or State Government. As a result, Phase 1 - Initiation is often overlooked due to time constraints and a desire to 'get on with the project', effective project managers will resist this pressure.

Achieving clarity in the early stages of the project is crucial for later success. If the project is unfeasibly defined and scoped, and not properly linked with Council's organisational goals and objectives, it will be difficult to obtain agreement amongst stakeholders and the project is unlikely to be completed successfully.

4. Governance

Project governance refers to the process by which the project is directed, controlled and held to account. The aim of project governance is to plan and manage the project throughout its life in order to achieve success. This is the time to make sure you know who is responsible for the project, who are the decision makers and which council officers have responsibility for certain aspects of the project. For example, the council may have personnel in charge of assessing risk and these people need to be involved in conducting the project risk assessment.

Governance should clearly define delegations and who can make decisions and commit funds. It should also include what decisions need to be referred to executive committees, full council, mangers and funding bodies

You also need to clearly clarify your own role in relation to the project and what decision -making authority you have as this can be different from your usual operational responsibilities.

The project's governance structure should be made clear including how it will operate within the general management structure of the council. Even though the structure may be the same or similar, with the same people, the distinction between the project activities (managed through the project governance structure) and normal ongoing business activities should be made clear. Blurring operational and project roles and reporting hierarchies has the potential to:

- negatively influence the project scope by causing confusion about objectives and what is to be done and delivered to whom
- create stress among the Project Team
- confuse reporting responsibilities and authority for project related decisions; and/or
- compromise the successful realisation of the project outcomes and business benefits in the longer term.

5. Stakeholder Engagement

Stakeholder engagement is the process of identifying the key stakeholders, analysing their influence on the project and managing their impact - including gaining their support where possible.

Stakeholders can be defined as key or non-key for the purpose of planning engagement strategies.

Key stakeholders are those individuals or groups whose interest in the project must be recognised if the project is to be successful, those stakeholders will be positively or negatively affected during the project or on successful completion of the project. Non-key stakeholders are those individuals or groups identified as having a stake in the project but who don't necessarily influence its outcome.

First start by identifying internal and external parties. Internal parties include Council, Divisions/Departments and employees. External parties include the community (residents, businesses/groups, workers, students and visitors), other levels of Government (State and Federal) and the media. It should be noted that some people may be members of more than one stakeholder group.

Where a project is funded by the NSW Government, they become a primary stakeholder, and personnel from the agency responsible for the funding will form part of the project team. It is essential to keep them informed of progress on a regular basis and flag issues early.

It is important to develop an understanding of the needs, values and issues that stakeholders have and address them to keep stakeholders engaged for the duration of the project. Early engagement with all stakeholders is essential to reveal their various expectations and assumptions, to manage their level of input and temper expectations about their level of influence. This is important for all projects, but especially important when the project is large, highly visible, political and/or business critical.

When a project has support across the organisation or within a community, the recruitment of 'champions' (i.e. project supporters who can motivate others either by role/status or personality or both) to openly promote and defend the project can be an important way to lend the project authority and build wider support among stakeholders.

Council may have a communication team or community engagement personnel who can assist or conduct community consultation. Involve these people early in the project as community consultation can be time consuming and needs to be built into the project timeline.

6. Gaining Approvals

It is essential to factor into your project the time needed to gain approvals. These approvals may include:

- Permission from the landowner
- Planning approval from government
- Regulatory requirements specific to infrastructure projects in NSW
- Environmental approvals (EPA)
- Heritage approval and approval from state heritage agencies
- Indigenous communities

One of the biggest causes of delay to most projects is a failure to factor in the time needed for approvals. A project manager should allow plenty of time to gain approvals and factor it into the scope and time frame.

7. Risk Management

The management of risk is quite straightforward. You need only ask yourself and your team one simple question: What can go wrong?

You need to ask the question with respect to all areas and dimensions of the project. Whether in the context of local government or anywhere else for that matter, these factors would include finance, quality, the availability of all types of resources, skill and performance, technical issues, stakeholder support, political intrusion, team dynamics, attitude problems, changing priorities and the effects of other projects and programs.

You could almost summarise the entire approach to risk as one aimed at eliminating or at least minimising the element of surprise. You never want to be surprised because it implies that you are not fully in control. If surprise elimination sounds ambitious, perhaps you would accept, at the very least, the minimisation of the element of surprise. How can you accomplish this?

There are always risks associated with a project. Successful project managers try to resolve risks before they occur, through a systematic risk management process.

Risk management ensures levels of risk and uncertainty are identified and managed in a structured way so any potential threat to the delivery of outputs (level of resourcing, time, cost and quality) and the realisation of project outcomes is appropriately managed, to ensure the project is completed successfully.

The first step is to identify the risk and then quantify it in terms of the likelihood of it occurring and the impact on the project if it does. You can then put strategies in place to eliminate or manage these risks. These strategies may include budgeting additional resources in case they are needed or revising the scope of the project.

Risks should be reviewed regularly throughout the project to ensure that changing circumstances are tracked and managed. Structured, proactive risk management allows risks to be anticipated and the effects minimised rather than taking a reactive approach to events as or after they occur, which can be costly.

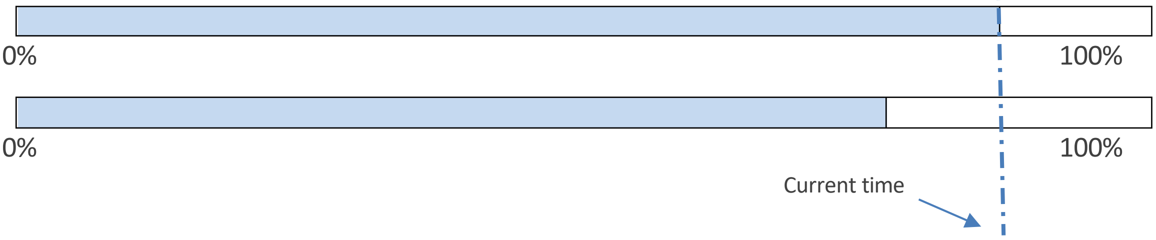
Effective risk management results in better service delivery, more efficient use of resources, and better project management, as well as helping to minimise waste, fraud and poor value for money decision making. Council's risk manager will be an essential part of your risk management.

Phase 2 Planning

Phase 2 involves creating a suite of planning documents (a Project Plan at a minimum) to organise the resources required to produce the project outputs and actions required to manage project risks, and to help guide the project through the remaining phases of the project. It generally expands on your project initiation documents. The project plan will also provide the dates and milestones for reporting.

When project planning, thinking into the future is needed. This includes identifying your assumptions, constraints and documenting the risks these pose to the projects. You can then add contingency measures to your project plan. In other words, you have thought of what to do for those 'what if this happened' scenarios.

Regardless of the project size, the most important document created during Phase 2 – Planning, is the Project Plan. This document is referred to frequently through Phase 3 - Delivery and Monitoring to ensure the project is on track. A well-defined Project Plan will ensure that the Project Manager has a clear view of the activities and milestones required to meet the customer's expectations. This is illustrated in the diagram below and it is easy to see the project is behind schedule.



When undertaking minor projects, these plans can be combined into a single document for approval by the Project Sponsor. The Project Manager will need to create more complex documents such as advanced risk assessments and financial models to ensure that the project activities are properly sequenced.

Information contained within the Project Plan must be consistent with the approved Business Case or funding agreement. Any major departure (variation) from the Project Plan must be negotiated with the Project Sponsor and/or General Manager and this may result in a requirement to resubmit a new Business Case or a funding agreement variation to the funding body. Good project planning should reduce, if not eliminate the need for funding agreement variation.

Most of the information you need will have been collected during the initiation phase but will be expanded in the Project Plan.

Actions – What to Do

Use the Project Planning Checklist (Checklist 2) and the planning tools to ensure you have a clear picture of the project and what it aims to achieve before moving onto Phase 3 Delivery and Monitoring. Discuss with the Project Sponsor how detailed to make the plan and choose a relevant template.

You should complete the following:

- Risk Assessment
- Detailed Project Plan and Scheduling Documents (Gantt chart or similar)
- Detailed Budget
- Communication Plan including meeting schedule and stakeholder consultation

Key Elements in the Planning Phase

1. Determining Risks

One of the crucial roles in project planning is to identify the project risks. These might include listing the known constraints, likely problems, or other possible issues & opportunities that might arise during the project. To create the initial list it is worth thinking about the things that have happened on other Council projects that will need to be well managed to ensure this project is successful. It is also worth thinking about opportunities that might arise if certain positive things eventuate and how you might plan for them in advance.

If you haven't run a similar project, discuss possible risks with someone with more experience or get a team together to brain-storm ideas.

Spending time identifying potential risks at this stage will allow you to consider how you might deal with (mitigate) these risks if they eventuate. It also enables you to discuss those potential issues with the Project Sponsor so that they are aware of what might go wrong; or where there might be opportunities to get additional benefits from the project.

Knowing what the potential risks are and how you might deal with them is vital to be able to carry out the next steps - namely to schedule the project, estimate costs, and determine the number & type of resources you will need to deliver the project.

2. Scheduling

In project management, the project schedule is a document that, if properly prepared, is usable for planning, execution, monitoring/controlling, and communicating the delivery of the scope to the stakeholders.

The main purpose of project scheduling is to represent the plan to deliver the project scope over time. A project schedule, in its simplest form, could be a chart of work elements with associated schedule dates of when work will take place and milestones (usually the completion of a deliverable) are planned to occur. In addition to guiding the work, the project schedule is used to communicate to all stakeholders when certain work elements and project events are expected to be accomplished. The project schedule is also the tool that links the project elements of work to the resources needed to accomplish that work.

At a minimum, the project schedule should include the following components:

- All key activities
- A planned start date for the project
- Planned start dates for each activity
- Planned finish dates for each activity
- Planned finish date for the project
- Milestones such as due dates for reports
- Resource assignments
- The "flow" (sequence) of the various activities
- The relationships of activities (the order in which work is done)

To develop project schedule list all the activities based on your work breakdown structure and sequence these. Identify the critical ones and add the milestones. Estimate the time needed to complete the activities and which ones can be done at the same time. Remember you may need to complete one or more activities before you move onto another one. You can then estimate the resources you will need to complete each activity and start getting these organised. This may require outsourcing, hiring equipment or scheduling work rosters.

There are three main types of schedules:

Master project schedule. A master schedule tends to be a simplified list of tasks with a timeline or project calendar.

Milestone schedule or summary schedule. This type of schedule tracks major milestones and key deliverables, but not every task required to complete the project.

A detailed project schedule. This is the most thorough project schedule, as it identifies and tracks every project activity. If you have a complex, large, or lengthy project, it's important to have a detailed project schedule to help track everything.

The most common form of project schedule is a Gantt chart. Both a milestone schedule and a detailed project schedule can be created as a Gantt chart. When choosing a scheduling softwar e, look for scheduling tools that allow you to create different views from the same schedule. If you create a detailed schedule with milestones as a Gantt Chart, make sure it can be summarized up to that level for a simpler view that can be easily shared with your team or stakeholders. This gives you the ability to present the same schedule in different formats depending on the level of detail required and the target audience.

Project scheduling provides the following benefits:

- Assists with tracking, reporting on, and communicating progress
- Ensures everyone is on the same page as far as tasks, dependencies, and deadlines
- Helps highlight issues and concerns, such as a lack of resources
- Helps identify task relationships
- Can be used to monitor progress and identify issues early.

Tips for creating a project schedule

Keep these seven tips in mind to make sure your schedule is realistic.

1. Get input from stakeholders. Make sure you don't create your schedule in isolation. It's important to use your team and other stakeholders to identify tasks, resources, dependencies, and durations.
2. Reference past projects. Looking at previous projects with similar scope and requirements can help create realistic estimates and ensure you haven't forgotten any tasks.
3. Include project milestones. Milestones are events or markers that stand for an important point in your project. They're useful for creating a summary schedule, reporting to executives, and identifying problems early. Here are some milestone examples:
    - Project kickoff
    - Design approvals
    - Completion of requirements (eg. funding deadlines, financial year, other delivery constraints like school holidays, etc.)
    - Product implementation

- Project closeout

4. Consider any non-work time. For example, make sure vacations and holidays are reflected in your schedule so that you're not assuming people will be working when they're not.
5. Define the critical path on your project. Identifying your project's critical path allows you to prioritize and allocate resources to the most important tasks in the project.
6. Record scheduling assumptions. Write down the logic behind your scheduling predictions. For example, if you assume it will only take 10 hours to complete a task because you're going to have a senior engineer. That way, if you end up with a junior engineer, you can understand and explain why it took twice as long as planned.
7. Keep risk in mind. Identify and document any factors that pose a risk to staying on schedule. This will help your risk management efforts.

3. Cost Estimation

Resource management relates to the financial, human, physical and information resources required to deliver a project, regardless of project category or size. Planning how to manage these resources is vital and will ensure resources are better managed and provide transparency for key stakeholders.

During Phase 2 Planning, a detailed project budget should be developed that reflects the resources required to complete the activities and tasks of the project, this should include any staff costs, training costs, physical resources, services or consultancies necessary to undertake the project and/or project management costs. Also allowances for contingencies associated with risk management strategies. For third party suppliers, the requirements of those suppliers will be defined within the relevant contract.

This should include an estimate of the financial contribution (real or notional) made by another organisation (e.g. project partner/grant funding) to provide an accurate cumulative total cost. The funding sources should be detailed in the Project Plan.

Once the individual costs have been estimated and linked to project activities or milestones, an overall project budget can be developed, including budget phasing-allocating funds when needed at various points in the project.

Do not rely on the budget from a similar project or on costs and estimates used in the past. Always check that these have not changed or where savings may be able to be applied.

Council's Financial Services and/or Procurement Services are able to assist with cost estimating, sourcing resources and budgeting.

Any changes to the initial budget should be documented (including budget variations), as should any issues that arise as a result of the budget or funding agreement, for reference during the evaluation process. At the end of the project it may be necessary to consider what should happen to any excess funds.

4. Resourcing and Procurement

During the planning phase of the project, it is essential to include planning for procurement of resources, including the engagement of contractors. In most councils there will be a finance department or office who can assist with the procurement of goods and services and the development of contracts.

It is important to have contracts drawn that work in with your project milestone dates to minimise the risk of delays and scope creep. The contracts may have penalties for failure to deliver to a time schedule. Consideration should be given to fixed pricing to better manage the project budget.

There may also be a funding agreement with a funding body such as the NSW Government. This contract should also inform your project planning and it is essential to read the contact carefully and be aware of the agreed deliverables and budget. If these aspects need to be negotiated this is best done prior to signing the contract.

Don't forget that some internal Council resources are also likely to be needed at some stage of the project. Even if staff involvement is only minor, the Project Manager needs to identify who, when and how staff will be involved and confirm that those resources will be available when required (eg according to the project schedule). Remember that staff will "have a day job" and will have to find "spare" time to help on the project.

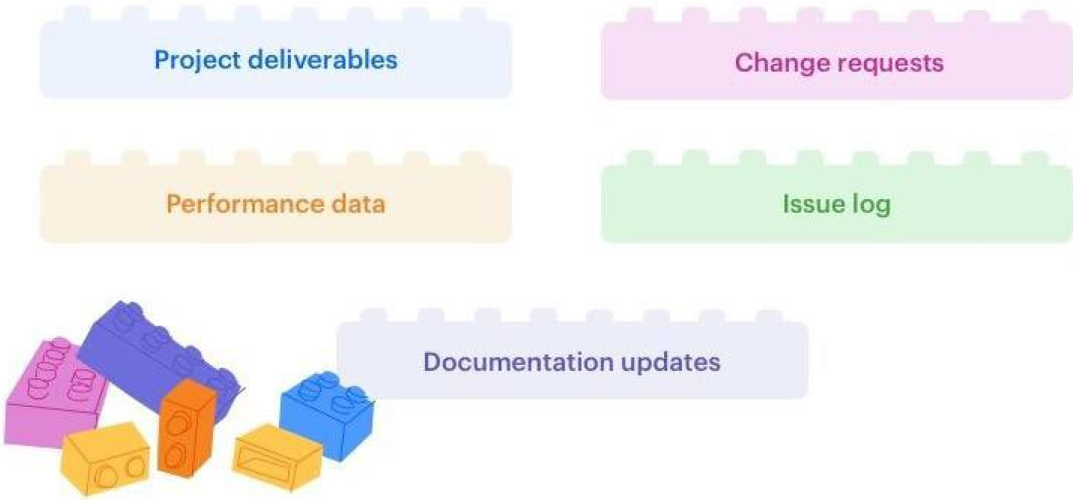
The greater the number and time staff will be involved, the greater the importance of this coordination step. It may also be necessary to discuss the need for internal resources with supervisors or managers of those staff to ensure they are supportive.

Phase 3 Delivery

In the delivery (execution) phase the team carries out all the planned activities during this stage. In project management, there might be hiccups along the way but if you catch them early on, it's easy to course-correct. You'll need to continuously track the project's progress and ensure that the milestones and deliverables stick to the project schedule. For this reason, the execution stage always includes project *controlling and monitoring.*

The project manager needs to be communicating with those delivering the project on a regular basis, reviewing the scope and risk documents and reporting on progress to stakeholders. At this stage the drivers of time and costs come together to create project quality. A quality project meets its deliverables within time and cost.

WHAT'S PRODUCED DURING PROJECT DELIVERY?



Depending on the nature of the project and council policies, you'll decide the sequence of activities that will happen during the delivery phase but overall:

- Execute the project scope
- Manage the team's work
- Recommend changes and corrective actions (remember early and clear communication of good and bad news is absolutely necessary)
- Manage project communication with stakeholders
- Celebrate project milestones and motivate team members
- Hold review meetings to make sure everything is on schedule
- Document all changes to the project plan, scope and costs.

There's no way to know if you're staying on track if you aren't measuring the project's progress. Your project planning process included setting measurable goals, deliverables and milestones. This is where all the effort you put into documentation comes in handy.

During the delivery phase, risks may materialise. By continuously assessing the risks, you equip your team with contingencies and keep the project from failure.

Almost every project suffers from a nasty monster called *scope creep*. It is when the project slowly grows out of your control and beyond the project's original scope. It generally includes blow outs in time and cost. If caught quickly these can generally be managed but you may need to consider contract variations as a last resort. Recording change requests, having them approved and communicating the change to the team and stakeholders is a main focus at this stage of the project. Contractors will also have conditions in their contracts and these may have to be actioned if they are not delivering as required.

Now, you cannot and should not avoid all change requests. Most projects require you to adapt to change. Some of these you should have anticipated in the planning stage, such as weather delays and have contingencies in place ready to implement.

PROJECT DELIVERY STEPS

Actions – What to Do

Use the Project Delivery Checklist (Checklist 3) and the delivery tools to monitor and record the delivery of your project deliverables before moving onto Phase 4 Close Out.

You should complete the following:

- Change Request Form
- Project Team Meeting Agenda
- Project Issues Register
- Project Status Report or other report such as
    - Report on a Page
    - Traffic Light Report
    - Dashboard
- Record of Stakeholder Feedback and Engagement (meeting minutes)

Key Elements in the Delivery Stage

1. Issues Management

Projects of any size have issues to deal with. Issues can be raised by anyone involved with a project. An issue is a concern that may impede the progress of a project if it is not resolved. Issues management involves monitoring, reviewing and addressing issues or concerns as they arise through the life cycle of a project.

Issues management is an essential skill for project managers. If issues are not addressed/resolved, they may become a risk to the project. Issues must be resolved quickly and effectively, and in some cases may require the involvement of the Project Sponsor and the General Manager. Depending on the nature of the issue, appropriate information should be reported regularly to management throughout the project. The Project Manager's role is to manage issues (and risks and changes). However, when they become more than minor, the PM needs to communicate what the impacts will be on the project (eg costs, timing, quality, etc) and gain Sponsor agreement on the best course of action.

Having a process to identify issues may reveal specific triggers that help to identify major risks before they occur. For example, continual rescheduling of team meetings may indicate that members are disengaged. This can seriously impact upon the project, as decisions will be delayed, and the process will stall. For minor projects a brief scan and ongoing monitoring may be all that is required to monitor issues, with identified issues documented in a table or spreadsheet. For major projects, it is advisable to maintain a project issues register, this should reflect the issue, current status and resolution.

2. Quality Management

The purpose of quality management is to ensure that the project management processes are conducted in a quality manner (quality assurance) and that outputs are delivered fit-for-purpose according to agreed quality criteria (quality control).

If quality management is not incorporated into the project management processes, it is probable that project outputs may not be fit-for-purpose and, subsequently, planned project outcomes will not be realised or will be realised to a lesser extent.

Quality management reduces the risk of project failure. This includes a process for managing changes, problems, issues and incidents that emerge during the project lifecycle and the production of the outputs. Quality management procedures need to be thoroughly planned for by the Project Manager.

For minor projects these procedures may not be formalised but should be scanned and ongoing monitoring may be all that is required to monitor issues, with identified issues documented in a table or spreadsheet.

For major projects, it is advisable to maintain a project issues register, this should reflect the issue, current status and resolution.

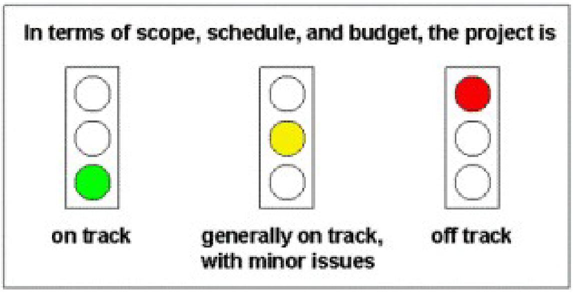
3. Reporting

Project status reporting is regular, reporting on the progress against the Project Plan. Usually by the Project Manager to the Project Sponsor and relevant managers, depending on the size, complexity and governance structure of the project. These reports may also go the council.

Project status reporting should highlight any problems that are occurring or have the potential to occur.

A document prepared for the **Safe and Secure Water Program** is included in the template pack as a reference for the reporting and quality assurance of NSW Government.

Reporting against agreed parameters or indicators can take the form of 'traffic lights' i.e. specific information is collated and assessed against the agreed parameters or indicators. The resulting status of 'health' of the project (or specific project components) is shown visually with an indicative traffic light colour. Green for on track/completed, yellow for progressing with minor issues and red for at risk/off track.



The frequency of reporting will vary depending on the size of the project and the requirement form its Project Sponsor. The project plan should include reporting dates as milestones.

Reporting should occur often enough that progress could be reported against several milestones since the last meeting/report.

Minor projects may only require infrequent consideration by the Project Manager and/or a meeting with the Project Sponsor to discuss any issues that could affect progress. The traffic light approach or report on a page may be sufficient for a minor project with low risk

For major projects the Project Status Report forms and integral part of the project, as information for the reports is drawn from the project management processes in place for the project.

4. Reviewing Scope and Risk

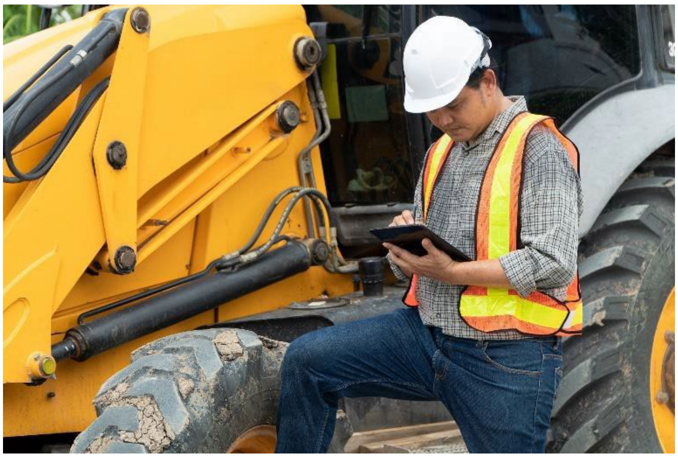
It is essential to continually undertake risk assessments through the life of the project. Reviewing and revising the risk assessment undertaken in the initiation phase should be undertaken at least weekly and more often on high-risk projects. The complexity of a project is measured not only in its monetary value but by the level of risk involved.

Reviewing the scope is also essential to ensure that tasks are not being added to the project that were not part of the original planning. If the scope changes this will require a review of plans and costs

5. Managing Contracts

The majority of council projects will involve contracts of some kind. These may be with the funding supplier (Federal or State Government), sub-contractors and suppliers of goods and other services.

It is essential that these contracts are well set up in the initiation stage and regularly reviewed during delivery of the project. The project manager needs to be aware early of when a contractual arrangement is at risk of not being met. This could be a delay in supplies or the failure of a sub-contractor to complete work on time.



If contract penalties need to be enforced or variations with funding bodies need to be made these should be done in a timely fashion and not at the last minute. Good communication, setting clear expectations and regular monitoring of progress are essential in ensuring contractual obligations are met.

6. Registering Changes

Change is a natural occurrence when managing a project. It is not the change itself that creates an issue, it is how the impact of the change is controlled and managed that is important. This includes identifying changes quickly, recording them and acting on them.

In project management, change management refers to the process used to identify, document, and remediate change in a project. That change could alter the scope, budget, resourcing, and timeline of a project. Or it could just alter an existing project requirement and nothing else.

Every organization handles change management differently, but a change request form is a simple tool you can use to document and track ongoing change. Changes can be recorded in the project change log and this provides excellent information when reviewing the project and documenting lessons learned.

Whether you have a change management process in place or not, it's important to think through the logical steps you might take to accept and agree to a project change.

Assess the impact of the change request

When a stakeholder or team member makes a change request, you'll want to inspect it to ensure the change is necessary, then assess the impacts.

Here are a few simple questions to help you and your team determine how to handle the change:

- What is the change?
- Why is the change being introduced?
- Does the change contribute to our project goals?
- What challenges does the change present?

Once you've discussed these questions with your team and stakeholders, you can focus in on how the change will impact your timeline and budget. It's a good idea to get your team together to talk about the level of effort required by the impending change.

For instance, let's say you're scoped to install a new pool at the swimming centre and the council decides to add water slide after the pool has already been dug. You'll need to figure out how that affects the existing plan, labour, and, eventually, costs before moving forward. After all, stopping work to redesign the pool, get the approval on the new design, and order the pro per materials could be expensive and time-consuming.

In some cases, a stakeholder may be fine with the change and its associated costs. But often, it can be just as much of a surprise to them as the original change request.

Be thoughtful, and spell out all the steps the change will require—making sure to get specific about potential project schedule and budget impacts. That way, everyone can agree on next steps and any added costs before you proceed with the work.

Adjust your project plan

Once everyone agrees on next steps, it's time to update your plan. If the change affects your timeline, be sure to capture a baseline of your plan before you reschedule tasks so you can track changes to your plan over time. You can use those learnings to inform project decisions and improve future planning. Then adjust all impacted tasks in your plan to ensure it is up-to-date

Of course, anytime you make changes to your plan, you'll need to let your team and stakeholders know. That way everyone's on the same page about the path moving forward and can work together to bring the project home successfully.

Here are a few simple ways you can communicate project changes:

- Include the updates in your next project status report.
- Note any timing, scope, or staffing changes at a team meeting and record in the minutes
- Email a view-only link to your plan to leaders and stakeholders with a brief explanation of what's changed.

Project Change Order Request

Project name:

Requested by:	**Date:**

Request name:	**Request number:**

Change description:

Change reason:

Impact of change:

- **Scope**:
- **Budget**:
- **Timeline**:
- **Resourcing**:
- **Communications**:
- **Other**:

Proposed action:

Associated cost:

Approved by:	**Date:**

Phase 4 Project Close

So much time and effort are put into the planning of a project, it is often forgotten that the end of a project is equally important. There's a lot of work involved even once a project is technically complete. The tasks might be procedural, but that doesn't make them any less important. There are sign-offs, signatures, payments, contractual reports and financials, as-built information, operational maintenance manuals, asset management records to be finalised. Don't forget to acknowledge the project success in some way.

If a project is abandoned, placed on hold or not realised a project close out should still be done. There are often good reasons why a project may not be completed and the decision not to proceed may be the best course of action, especially where considerable risks present themselves after project commencement.

Evaluating a project after it has been formally closed is an opportunity to assess how the project performed, whether success was achieved, how the changing expectations were managed and what lessons were learned. Post project evaluation should be seen as an opportunity to acknowledge what went right with the project and identify what could have been done better without attributing blame to individuals.

It should be viewed as an opportunity to contribute to the continuous improvement of the quality of project planning and implementation within the organisation. Realistic information on the project team's performance, the stakeholders' contribution, the accuracy of estimates and the responsiveness of the project to changes can all be assessed with a view to preparing recommendations for other projects to prevent similar mistakes from occurring.

Experience has shown that this stage is usually done badly or not at all, typically because of perceptions that an evaluation is too time consuming, or there is a 'lack of interest' by key players in the project governance structure. Often there is a rush to move onto the next project.

Before a project can be evaluated, specific agreed criteria are required as a basis for assessing success, including:

- determining whether key performance milestones have been met
- how well the project was managed
- budget and cost control
- whether the specified project outputs have been delivered to the required quality and
- to what extent the project outcomes have been secured or realised.

The criteria for judging a project's success should be clearly documented in the project plan and should define the circumstances and timing of any post project evaluation. A project can be considered successful if:

- project objective(s) have been realised
- target outcomes have been achieved
- project outputs are delivered on time and to the agreed quality
- costs are within those budgeted
- risks were managed appropriately
- processes and methodologies were implemented appropriately
- stakeholders' requirements have been met; and
- lessons have been learned and documented for next time.

Key Elements in the Close Out Stage

- Start at the beginning with the project scope document you created and make sure that you've meet all the requirements listed there. Record issues and improvements as you work through the project to ensure these are not missed in the close out stage.
- Make sure that all deliverables have been handed off and signed by stakeholders, getting their approval and satisfaction. Gather asset data together and register within the enterprise asset management system.
- Other project documents must also be signed by the appropriate person, this includes any outstanding contracts and agreements with vendors and other contractors.
- Once documents are signed off on, then process them and pay off all invoices and close out any project-related contracts.
- Finalise all project reporting and contractual requirements
- Add all documents together, then organise and archive them as historical data to be used for future reference.
- Use collected paperwork to identify and document the lessons learned over the course of the project, including any feedback from stakeholders, so others have reference points for future projects.
- Convene a project team meeting to invite feedback about what worked, and what didn't. Encourage honesty. By documenting the mistakes and the successes of the project, you're building a catalogue that offers historic data. You can go back and look over the information for precedents when planning for new projects.

Actions – What to Do

Use the Project Close Checklist (Checklist 4) and the closure tools to monitor and record the close of your project.

You should complete the following:

- Project Closure Checklist
- Lessons Learned Summary
- Final Report to Project Sponsors and council

Conclusion

Successful projects use a systematic approach, with planning and monitoring at every stage. There are many reasons why projects fail, and every organisation has examples of projects that can be considered failures. Many project failures could have been avoided through appropriate use of project management tools and techniques. The most commonly cited reasons for project failure, in no particular order, are:

- poor or no relationship to the organisation's strategic priorities
- lack of feasibility, including poor business case, poor estimates, decisions based on price rather than long term value for money, unrealistic timelines
- poorly articulated project objective(s) and project outcomes with unachievable and/or unverifiable targets
- poor or inadequate project scoping, poor scope management, including poorly defined and/ or documented scope, no periodic review of business needs to revalidate the project scope, over ambitious planning (e.g. development and implementation not broken down into manageable steps), poor tracking of scope changes (e.g. poor or absent document approval processes), inadequate understanding of the impact of changes to the scope, unwillingness to review and adjust initial estimates to account for normal cost increases over the longer term
- inadequate governance, including poor ownership (e.g. disengaged Project Sponsor) and passive leadership from management executive, lack of stability in project governance, lack of clarity in roles, no acceptance of responsibility and accountability, no transparency in decision making, irregular and/or informal status meetings, poor engagement in the decision making processes (e.g. unwilling to make decisions or be accountable)
- inadequate record keeping, including gaps in critical documentation, undated and/or unsigned records, lack of author details or cross references to source documentation;
- poor resourcing or contract management;
- poor management of change, including lack of management/stakeholder buy in to the change, poor or no planned approach to managing the internal business change required for successful implementation
- poor stakeholder engagement, including failure to set and manage expectations, poor and/or infrequent communication (e.g. mixed messages, messages do not reflect the project stage, messages not tailored for stakeholder groups), no process to resolve stakeholder issues
- poor expectation management, including public expectation
- inadequate risk management, including poor risk identification, failure to monitor or reassess risks, failure to identify emerging risks and apply mitigation, no contingency funding, and responsibility for risk mitigation not allocated and/or not accepted; and
- inadequately trained and/or inexperienced project managers.

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Glossary of Common Project Management Terms

Activity	A task, job, operation or process consuming time and possibly other resources. (2) The smallest self-contained unit of work in a project.
Activity duration	The length of time that it takes to complete an activity.
Actual dates	The dates on which activities started and finished as opposed to planned or forecast dates.
Actual expenditure	The costs that have been charged to the budget and for which payment has been made or accrued.
Business Case	Provides justification for undertaking a project, programme or portfolio. It evaluates the benefit, cost and risk of alternative options and provides a rationale for the preferred solution.
Change Log or Change Register	A record of all project changes: proposed, authorised, rejected or deferred.
Closeout	The process of finalising all project matters, carrying out final project reviews, archiving project information and redeploying the remaining project team.
Closure	The formal end point of a project, programme or portfolio; either because planned work has been completed or because it has been terminated early.
Contingency budget	The amount of money required to implement a contingency plan.
Contingency plan	Resource set aside for responding to identified risk.
Cost estimating	The process of predicting the costs of a project.
Dependencies	Dependencies specify the relationship between project activities and the order in which they are to be performed. There are 4 kinds of dependencies Start-to-start – Predecessor task must start before the Successor can start Finish-to-start – Predecessor must finish before Successor can start Start-to-finish – Predecessor must start before Successor can finish Finish-to-finish – Predecessor must finish before Successor can finish.
Gantt chart	A graphical representation of activity against time.

Governance	The framework of authority and accountability that defines and controls the outputs, outcomes and benefits from projects, programmes and portfolios. The mechanism whereby the investing organisation exerts financial and technical control over the deployment of the work and the realisation of value.
Issue management	The comprehensive process of identifying, resolving, and tracking issues associated with your projects comes under issue management. The purpose of issue management is to timely resolve the issues before they become big disasters.
Issue tracking	Issue tracking is the process of identifying a possible bug or error in the product which is affecting its optimal performance. Most of the time, a professional issue tracking software is kept in place for efficient issue tracking.
Issue log/register	A complete record of all the project issues (ongoing and closed), along with the persons responsible for resolving them is included in an issue log. The document may also include each issue's status and resolution deadlines. This could be managed through the change log.
Key milestone	A milestone, the achievement of which is considered to be critical to the success of the project.
Kick-off meeting	A kick-off meeting is generally the first meeting that occurs between the project team and their client. This meeting usually occurs after the basic project details have been finalized, but the main project work has not been started yet. It serves the purpose of reviewing the project expectations and to create alignment between everyone involved in the project.
Meeting agenda	A meeting agenda is simply a list of all the topics that are to be discussed during a meeting. It may include detailed topic descriptions, their sequence, and the expected outcomes of each topic.
Meeting minutes	Meeting minutes are written notes of whatever is discussed during a meeting. These minutes can be circulated among meeting participants after the meeting to gain valuable insights and take appropriate follow-up actions.
Milestone	A milestone represents a major event in a project lifecycle. It is used as a reference point to measure the progress of a project. Usually represented as diamonds, milestones greatly help with project scheduling and monitoring.
Progress report	A regular report to senior personnel, sponsors or stakeholders summarising the progress of a project including key events, milestones, costs and other issues.
Project initiation document (PID	A document approved by the project board at project initiation that defines the terms of reference for the project.

Project brief	The project brief contains a brief description of the objective and background/context for the overall project, and outlines the project management framework to be adopted for the initiation phase, including how to measure the success of the phase; deliverables, budget and resources; project activities and milestones; governance and reporting requirements; communication with stakeholders; risk and issues management; assumptions and constraints; related projects and interdependencies; guidelines/standards; levels of review; quality assurance and capturing lessons learned.
Project budget	Project budget is a formally approved document featuring a comprehensive list of financial resources, including project expenses, required to complete a project.
Project charter	A project charter is a formal, typically short document that describes your project in its entirety — including what the objectives are, how it will be carried out, and who the stakeholders are. It is a crucial ingredient in planning the project because it is used throughout the project lifecycle.
Project manager	The person responsible for handling every aspect of a project from the day it starts till it closes is called a project manager. The responsibilities of a project manager typically entail powerful planning, smart resource utilization, and managing the scope of the project.
Project plan	A project plan is one of the key formal documents created before starting any project. The document usually consists of approved cost, schedule, and project scope. It guides the execution of a project from initiation to project closure. The project plan also lays the foundation for all kinds of communication among the stakeholders.
Project stakeholder	Any individual that has a direct or indirect interest in a project is known as a project stakeholder. They usually affect or are affected by the project decisions being taken over the course of the project lifecycle. A stakeholder can be anyone from the project team, executives, sponsors, customers, or the end-users.
Project sponsor	The project sponsor is an individual (often a manager or executive) with overall accountability for the project and acts as the link between the project, the stakeholders, and strategic level decision-making groups.
Project timeline	A project timeline outlines the project events in order of their occurrence. It captures exactly what needs to be done over the course of the project lifecycle and how it will be done.

Quality assurance	Quality assurance is a set of planned and systematic activities implemented to monitor the project processes in a way that project quality requirements are fulfilled. Quality assurance is done during the project and can involve regular quality audits.
Quality control	Quality control involves the use of standardized practices to evaluate whether the resulting product of a project meets quality expectations or not. The process is conducted after the product has been created to identify any changes that might be required in the quality assurance process.
Resource allocation	Resource allocation involves scheduling and assigning resources for a project in the most efficient way possible. The purpose of resource allocation is to maximize the use of available resources in a way that supports the project's end goals.
Resource breakdown structure	A comprehensive list of resources required to complete a project. This list is usually made according to the function and resource type, facilitating the planning and control of a project work.
Risk management	The process of identifying and assessing risks to decrease their negative impact on project operations is called risk management. During the process, it is made sure that overall project goals are not affected in any way.
Risk mitigation	A strategy devised to decrease the probability of adverse effects of risk is known as risk mitigation. A successful risk mitigation strategy focuses on developing actions that reduce possible threats to overall project objectives.
Risk monitoring and control	Risk monitoring and control involves tracking of how the risk responses are performing in comparison with the original risk management plan.
Risk owner	A person responsible for ensuring that a particular risk is managed appropriately is a risk owner. One of core duties of a risk owner is to make sure that the mitigation strategy is implemented effectively. He can also sometimes be involved with performing qualitative and quantitative risk analysis.
Scope	Project scope is the part of project planning that involves determining and documenting a list of specific project goals, deliverables, tasks, costs and deadlines. It defines what is covered by the project and what is not. The scope statement also provides the team with guidelines for making decisions about change requests during the project.
Scope creep	The term sometimes given to the continual extension of the scope of some projects.

Scope management	The process whereby outputs, outcomes and benefits are identified, defined and controlled.
Stakeholder grid	A matrix used as part of a stakeholder analysis to identify the relative importance of stakeholders to a project; for example, by considering their relative power.
Stakeholder identification	The process of identifying stakeholders in a project.
Work Breakdown Structure (WBS)	A work breakdown structure comprehensively divides the project deliverables into manageable sections. This hierarchical organizing of the team’s work helps everyone understand the nature of work better and execute project goals effectively.